

Financial Technology

Meeting #3

PHP Dunia Bayar Indonesia

Prof. Dr. Ridwan Sanjaya, MS.IEC

Scoring

- Assignments and Presentations 40%
 - Mid Term 20%
 - After Mid Term 20%
- Mid-Term Exam 30%
- Final Exam 30%

Konten

1. Ide Aplikasi Fintech
2. Pengenalan Postman
3. PHP Dunia Bayar Indonesia

Peringatan

API Dunia Bayar (AiYO) akan digunakan untuk level production dalam project Mata Kuliah Teknologi Keuangan (Fintech)

Sehingga pembayaran yang terjadi menggunakan API ini merupakan transaksi nyata. Gunakan nilai transaksi yang kecil untuk uji coba

AiYO

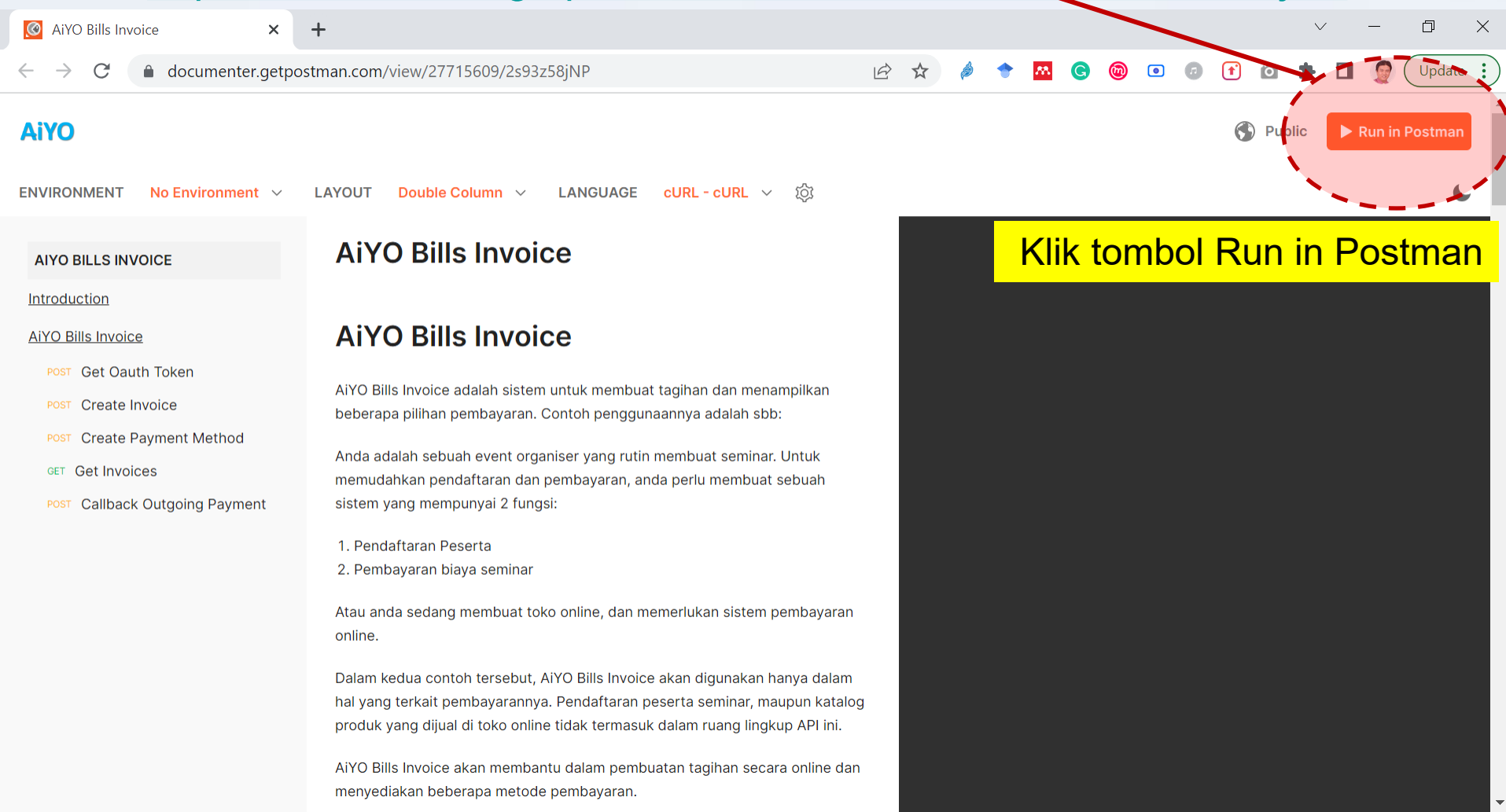
BILLS INVOICE

Akses URL API

- Untuk mendapatkan API-nya tuliskan URL <https://documenter.getpostman.com/view/27715609/2s93z58jNP> di browser web

AiYO Bills Invoice

- <https://documenter.getpostman.com/view/27715609/2s93z58jNP>



The screenshot shows a web browser displaying the AiYO Bills Invoice API documentation. The browser's address bar shows the URL <https://documenter.getpostman.com/view/27715609/2s93z58jNP>. The page title is "AiYO Bills Invoice". The left sidebar contains a list of API endpoints: "Introduction", "AiYO Bills Invoice", "POST Get Oauth Token", "POST Create Invoice", "POST Create Payment Method", "GET Get Invoices", and "POST Callback Outgoing Payment". The main content area is titled "AiYO Bills Invoice" and contains the following text:

AiYO Bills Invoice adalah sistem untuk membuat tagihan dan menampilkan beberapa pilihan pembayaran. Contoh penggunaannya adalah sbb:

Anda adalah sebuah event organiser yang rutin membuat seminar. Untuk memudahkan pendaftaran dan pembayaran, anda perlu membuat sebuah sistem yang mempunyai 2 fungsi:

1. Pendaftaran Peserta
2. Pembayaran biaya seminar

Atau anda sedang membuat toko online, dan memerlukan sistem pembayaran online.

Dalam kedua contoh tersebut, AiYO Bills Invoice akan digunakan hanya dalam hal yang terkait pembayarannya. Pendaftaran peserta seminar, maupun katalog produk yang dijual di toko online tidak termasuk dalam ruang lingkup API ini.

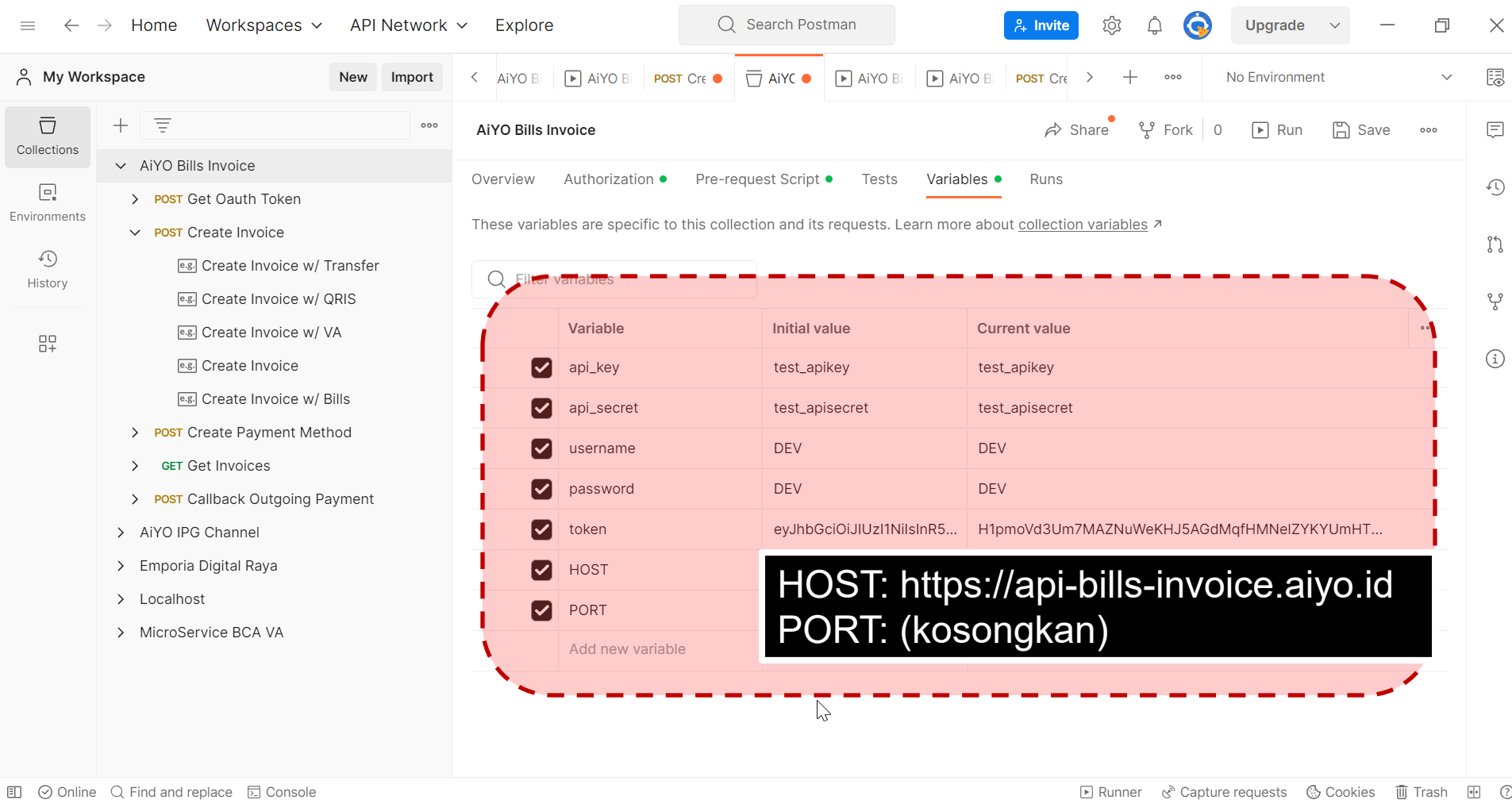
AiYO Bills Invoice akan membantu dalam pembuatan tagihan secara online dan menyediakan beberapa metode pembayaran.

In the top right corner of the browser window, there is a red button labeled "Run in Postman". A red arrow points from the URL in the list above to this button. The button is highlighted with a red dashed circle. A yellow box with the text "Klik tombol Run in Postman" is overlaid on the bottom right of the screenshot.

AiYO Bills Invoice

- Untuk membuat tagihan dan menampilkan beberapa pilihan pembayaran, contoh: pembayaran seminar, iuran, donasi
- Metode pembayaran
 - TRANSFER, bankCode: '014' → Create Invoice w/ Transfer
 - QRIS, bankCode: '503' → Create Invoice w/ QRIS
 - VA_CLOSED, bankCode: '002', '008', '009', '013', '014', '022', '484', '523' → Create Invoice w/ VA

Variabel



The screenshot shows the Postman interface with the 'AiYO Bills Invoice' collection selected. The 'Variables' tab is active, displaying a table of variables. A red dashed box highlights the table content. A black box with white text is overlaid on the bottom right of the table, providing the values for the 'HOST' and 'PORT' variables.

Variable	Initial value	Current value
☒ api_key	test_apikey	test_apikey
☒ api_secret	test_apisecret	test_apisecret
☒ username	DEV	DEV
☒ password	DEV	DEV
☒ token	eyJhbGciOiJIUzI1NiIsInR5cCI6Ikpz...	H1pmoVd3Um7MAZNuWeKHJ5AGdMqfHMNeIZYKYUmHT...
☒ HOST		https://api-bills-invoice.aiyo.id
☒ PORT		(kosongkan)
Add new variable		

API "AiYO Bills Invoice" dari DBI akan masuk ke dalam Workspace Postman

- Salin accessToken untuk digunakan pada aktivitas Create Invoice



Salin token ke variable

- Tempel accessToken ke variable token dan tekan tombol Save

The screenshot shows the Postman interface with the 'Variables' tab selected for the 'AiYO Bills Invoice' collection. The 'token' variable is highlighted with a red dashed box, and its value is being pasted into the 'Current value' field. A black box at the bottom contains the text 'HOST: https://api-bills-invoice.aiyo.id' and 'PORT: (kosongkan)'.

Variable	Initial value	Current value
☒ api_key	test_apikey	test_apikey
☒ api_secret	test_apisecret	test_apisecret
☒ username	DEV	DEV
☒ password	DEV	DEV
☒ token	eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZGVudGlmaW...	eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZGVudGlmaW...
☒ HOST	http://51.79.242.222	http://51.79.242.222
☒ PORT		
☒ public-token		
Add new variable		

HOST: https://api-bills-invoice.aiyo.id
PORT: (kosongkan)

POST | Create Invoice

- Jalankan API Create Invoice untuk menghasilkan tagihan

The screenshot displays the Postman interface for a REST client. The left sidebar shows the 'My Workspace' with a collection named 'AiYO Bills Invoice'. The main panel shows a POST request to the endpoint `{{(HOST)}}:{{(PORT)}}/api/v1/invoice`. The request body is a JSON object with the following fields:

```
{
  "invoiceName": "Tagihan 3 Barang",
  "referenceId": "RSG123123219",
  "userName": "Test AiYO",
  "userEmail": "aiyo@test.com",
  "userPhone": "0852123456789",
  "remarks": "Pembelian Barang Baru",
  "payAmount": "10000",
  "expireTime": "2024-12-28 18:00:00",
}
```

The response is shown in the bottom panel, indicating a successful status of 200 OK. The response body is a JSON object with the following fields:

```
{
  "responseCode": "2000000",
  "responseMessage": "Invoice Created",
  "responseData": {
    "invoiceId": "qdUaDCuxDo7wqVQWiu9I",
    "invoiceURL": "https://bills-invoice.aiyodev.xyz/bills/invoice/qdUaDCuxDo7wqVQWiu9I?accessToken=jD2bCLBFCWdndjigUVdNdnsVdCx5dXLEgsTXndTQ7v126vQcXX",
  }
}
```

The bottom status bar shows the following information: Online, Find and replace, Console, Runner, Capture requests, Cookies, Trash, and a question mark icon.

Contoh Body (raw)

```
{
  "invoiceName": "Tagihan 2 Barang",
  "referenceId": "RTD123123211291826",
  "userName": "Test AiYO",
  "userEmail": "aiyo@test.com",
  "userPhone": "0852123456789",
  "remarks": "Pembelian Barang Baru",
  "payAmount": "10000",
  "expireTime": "2024-12-28 18:00:00",
  "billMasterId": "eyyWmHQ65zBLulC1a86b",
  "paymentMethod": {
    "type": "VA_CLOSED",
    "bankCode": "022"
  },
  "items": [
    {
      "itemName": "iPhone 14 Pro Max",
      "itemType": "ITEM",
      "itemCount": "1",
      "itemTotalPrice": "5000"
    },
    {
      "itemName": "Laptop Gaming",
      "itemType": "ITEM",
      "itemCount": "1",
      "itemTotalPrice": "500"
    }
  ]
}
```

Salin invoiceURL

- Salin invoiceURL menampilkan tagihan

The screenshot shows the Postman interface with a workspace named "My Workspace". The left sidebar lists collections, environments, and history. The main panel displays a POST request to "AiYO Bills Invoice / Create Invoice" with a body containing invoice details. The response is shown in the "Body" tab, highlighting the "invoiceURL" field.

Request Details:

- Method: POST
- URL: `{{(HOST)}}:{{(PORT)}}/api/v1/invoice`
- Body (JSON):

```
{  "invoiceName": "Tagihan 3 Barang",  "referenceId": "RSG123123219",  "userName": "Test AiYO",  "userEmail": "aiyo@test.com",  "userPhone": "0852123456789",  "remarks": "Pembelian Barang Baru",  "payAmount": "10000",  "expireTime": "2024-12-28 18:00:00",}
```

Response Details:

- Status: 200 OK
- Time: 603 ms
- Size: 1.19 KB
- Body (JSON):

```
{  "responseCode": "2000000",  "responseMessage": "Invoice Created",  "responseData": {    "invoiceId": "qdUaDCuxDo7wqVQWiu9I",    "invoiceURL": "https://bills-invoice.aiyodev.xyz/bills/invoice/qdUaDCuxDo7wqVQWiu9I?accessToken=jD2bCLBFCWdndjigUVdNdnsvDcx5dXLEgsTXndTQ7v126vQcXX",  }}
```

The "invoiceURL" field in the response is highlighted with a red dashed box.

Tampilan Tagihan

- Jika invoiceURL disalin ke web browser, maka tampilan tagihannya

The screenshot shows a web browser window with the address bar displaying the URL: `bills-invoice.aiyodev.xyz/bills/invoice/IVrq2RMyp845AKxu2fUr?accessToken=H1pmoVd3Um7MAZNu...`. The browser's address bar also shows the page title "AiYO Bills Invoice - Detail Invoice".

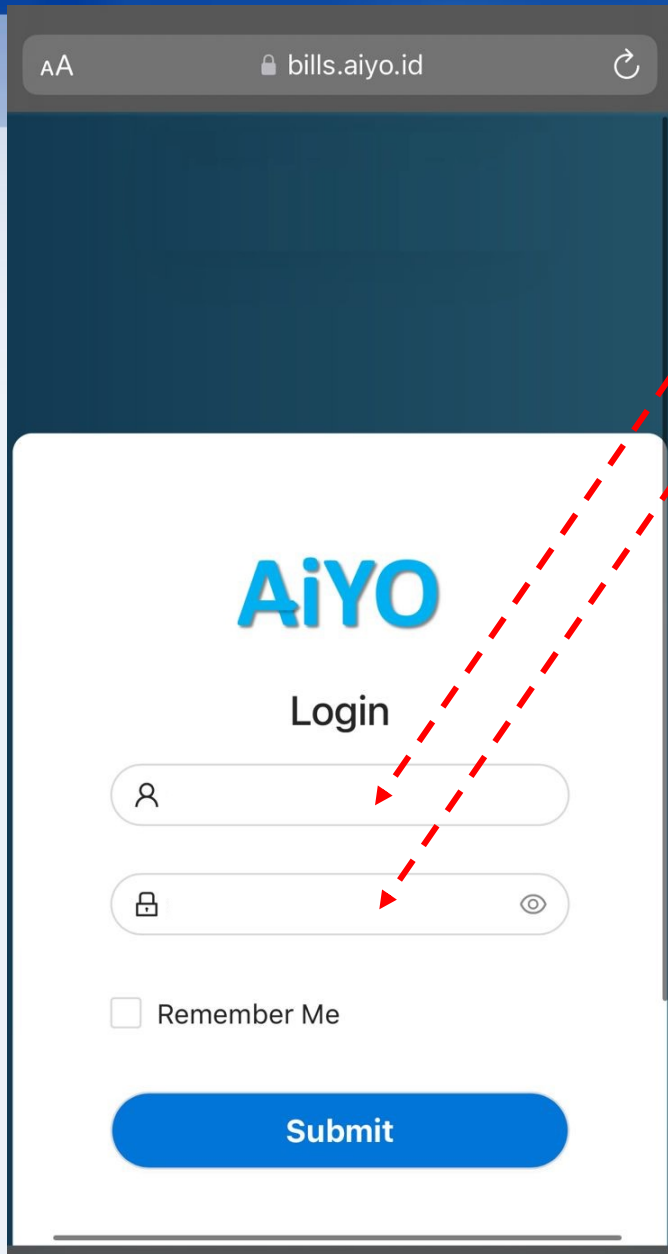
The main content area of the browser displays an invoice detail page for "AiYO". The page has a blue header with the text "Invoice Overview" and the "AiYO" logo. The main content area is white and contains the following information:

- Invoice For:** Tagihan 3 Barang
- Amount:** Rp. 10,000
- Invoice Expired In:** 512 Days 00:48:32
- Invoice Details:** (marked with a "NEW" badge)
- Reference ID:** RSA123123219
- Table:**

Name Test AiYO	Phone 0852123456789
Email aiyo@test.com	Invoice Date 4 Agustus 2023
- Note:** Pembelian Barang Baru

<https://bills.aiyo.id/>

- User sesuai email terdaftar
- Password



The image shows a web browser window with the address bar displaying "bills.aiyo.id". The page content features the "AiYO" logo in blue, followed by the word "Login". Below this are two input fields: the first for a username (indicated by a person icon) and the second for a password (indicated by a lock icon and a toggle eye icon). A "Remember Me" checkbox is located below the password field. At the bottom of the form is a blue "Submit" button. Two red dashed arrows originate from the bullet points in the adjacent list: one points to the username field and the other points to the password field.

AiYO

Login

☐ Remember Me

Submit

Contoh Dashboard DBI

Bill Master ID :	BglvzfXdTTgV80cBgMv7
Provider :	Tim Dana Keuskupan Agung Semarang
Name :	Opera Jenaka Romo 3
Bill Type :	One Time
Bill Date :	2000-01-01
Bill Due Date :	2099-01-01
Penalty Type :	No Penalty
Penalty Amount :	0
Admin Fee Type :	Include
Admin Fee :	500
* Minimum Admin Bill Amount :	500

Tabel transaksi

#	Name	Type	Collation	Attributes	Null	Default
1	referenceId	varchar(25)	latin1_swedish_ci		No	<i>None</i>
2	userName	varchar(50)	latin1_swedish_ci		No	<i>None</i>
3	userEmail	varchar(40)	latin1_swedish_ci		No	<i>None</i>
4	userPhone	varchar(20)	latin1_swedish_ci		No	<i>None</i>
5	remarks	text	latin1_swedish_ci		No	<i>None</i>
6	payAmount	bigint(20)		UNSIGNED	No	<i>None</i>
7	items	text	latin1_swedish_ci		No	<i>None</i>
8	invoiceId 	varchar(25)	latin1_swedish_ci		No	<i>None</i>
9	status	varchar(10)	latin1_swedish_ci		No	<i>None</i>
10	timestamp	timestamp			No	current_timestamp()

Tabel transaksi

```
CREATE TABLE `transaksi` (  
  `referenceId` varchar(25) NOT NULL,  
  `userName` varchar(50) NOT NULL,  
  `userEmail` varchar(40) NOT NULL,  
  `userPhone` varchar(20) NOT NULL,  
  `remarks` text NOT NULL,  
  `payAmount` bigint(20) UNSIGNED NOT NULL,  
  `items` text NOT NULL,  
  `invoicId` varchar(25) NOT NULL,  
  `status` varchar(10) NOT NULL,  
  `timestamp` timestamp NOT NULL DEFAULT current_timestamp()  
) ENGINE=InnoDB DEFAULT CHARSET=latin1 COLLATE=latin1_swedish_ci;
```

```
ALTER TABLE `transaksi`  
  ADD PRIMARY KEY (`invoicId`);  
COMMIT;
```

URL Callback

- <https://mesinbayar.com/app/gok/callback/>
- <https://mesinbayar.com/app/god/callback/>
- <https://mesinbayar.com/app/stenly/callback/>
- <https://mesinbayar.com/app/flowfi/callback/>
- <https://mesinbayar.com/app/zenta/callback/>
- <https://mesinbayar.com/app/msra/callback/>
- <https://mesinbayar.com/app/ps7/callback/>
- <https://mesinbayar.com/app/lbx/callback/>
- <https://mesinbayar.com/app/dem/callback/>
- <https://mesinbayar.com/app/fhk/callback/>

<https://app.mesinbayar.com/>

aiyo_config.php

```
<?
```

```
$host = "https://api-bills-invoice.aiyo.id";
```

```
$username = "xxx";
```

```
$password = "xxx";
```

```
$billMasterId = "xxx"; //lihat dashboard DBI
```

```
$api_key = "xxx";
```

```
$api_secret = "xxx";
```

```
?>
```

token.php

```
<?php
include "aiyo_config.php";

$pathToken = '/api/oauth/token';
$urlGetToken = $host . $pathToken;

$chGetToken = curl_init($urlGetToken);
curl_setopt($chGetToken, CURLOPT_RETURNTRANSFER, TRUE);
curl_setopt($chGetToken, CURLOPT_ENCODING, "");
curl_setopt($chGetToken, CURLOPT_MAXREDIRS, 10);
curl_setopt($chGetToken, CURLOPT_TIMEOUT, 0);
curl_setopt($chGetToken, CURLOPT_FOLLOWLOCATION, TRUE);
curl_setopt($chGetToken, CURLOPT_HTTP_VERSION, CURL_HTTP_VERSION_1_1);
curl_setopt($chGetToken, CURLOPT_CUSTOMREQUEST, 'POST');
curl_setopt($chGetToken, CURLOPT_HTTPHEADER, array(
    'Authorization: Basic '.base64_encode($username.':'.$password)
));

$result = curl_exec($chGetToken);
$nilai=json_decode($result);
$accessToken=$nilai->responseData->accessToken;
curl_close($chGetToken);
?>
```

respon.php

```
<?php
include "token.php";
//Create Invoice
$bodyCreateInvoice = array(
    "invoiceName" => "Nama Komunitas",
    "referenceId" => "YPD".date("mdHis"),
    "userName" => "Ridwan Sanjaya",
    "userEmail" => "ridwan@unika.ac.id",
    "userPhone" => "0818000000",
    "remarks" => "-",
    "payAmount" => 50000,
    "expireTime" => date("Y-m-d\TH:i", strtotime('+3 hour'));
    "billMasterId" => $billMasterId,
    "paymentMethod" => array(
        "type" => "VA_CLOSED",
        "bankCode" => "022"
    ),

    "items" => array(
        array(
            "itemName" => "Barang 1",
            "itemType" => "ITEM",
            "itemCount" => "1",
            "itemTotalPrice" => "10000",
        ),
        array(
            "itemName" => "Barang 2",
            "itemType" => "ITEM",
            "itemCount" => "2",
            "itemTotalPrice" => "20000",
        ),
    )
);
```

respon.php (2)

```
//Signature
$pathInvoice = '/api/v1/invoice';
$urlCreateInvoice = $host . $pathInvoice;
$signRelativeURLCreateInvoice = parse_url($urlCreateInvoice, PHP_URL_PATH);
$rawBodyCreateInvoice = json_encode($bodyCreateInvoice);
$dataToSignCreateInvoice = $api_key . $signRelativeURLCreateInvoice . $rawBodyCreateInvoice;
$signatureCreateInvoice = hash_hmac('sha256', $dataToSignCreateInvoice, $api_secret);
$chCreateInvoice = curl_init($urlCreateInvoice);
$headersCreateInvoice = array(
    "Content-Type: application/json",
    "Authorization: Bearer " . $accessToken,
    "x-aiyo-key: " . $api_key,
    "x-aiyo-signature: " . $signatureCreateInvoice
);
curl_setopt($chCreateInvoice, CURLOPT_TIMEOUT, 30);
curl_setopt($chCreateInvoice, CURLOPT_POST, 1);
curl_setopt($chCreateInvoice, CURLOPT_RETURNTRANSFER, TRUE);
curl_setopt($chCreateInvoice, CURLOPT_HTTPHEADER, $headersCreateInvoice);
curl_setopt($chCreateInvoice, CURLOPT_POSTFIELDS, $rawBodyCreateInvoice);
$sentHeaders = curl_getinfo($chCreateInvoice, CURLINFO_HEADER_OUT);
$responseCreateInvoice = curl_exec($chCreateInvoice);
```

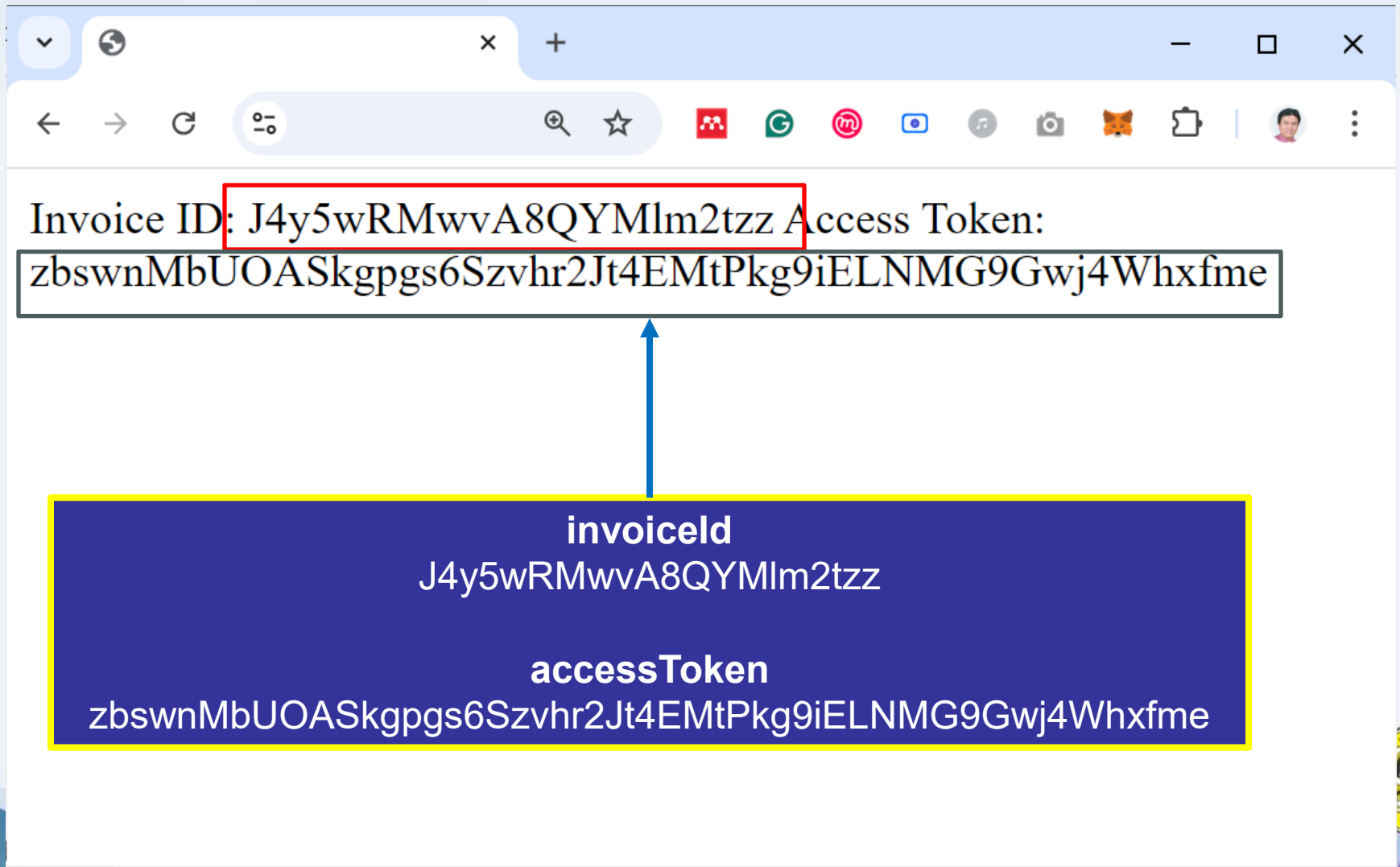
respon.php (3)

```
$invoice=json_decode($responseCreateInvoice);
if($invoice->responseCode=='2000000') {
    $invoiceId=$invoice->responseData->invoiceId;
    $accessToken=$invoice->responseData->accessToken;
    include "db_config.php";
    $sql = "INSERT INTO `transaksi` (`referenceld`, `userName`, `userEmail`, `userPhone`, `remarks`, `payAmount`,
`items`, `invoiceId`, `status`, `timestamp`) VALUES ('".$_POST['referenceld']."', '".$_POST['userName']."',
 '".$_POST['userEmail']."', '".$_POST['userPhone']."', '".$_str_replace( array( '\', \"', \"'\", \"'\", '<', '>', '/' ), '\"', $remarks).\"',
 '".$_POST['payAmount']."', '".$_detail.\"', '".$_invoiceId.\"', 'NEW', current_timestamp())";

    if ($conn->query($sql) === TRUE) {
        //echo "Data inserted successfully";
    } else {
        echo "Error: " . $sql . "<br>" . $conn->error;
    }

    // Close connection
    $conn->close();
}
curl_close($chCreateInvoice);
echo "Invoice ID: ". $invoiceId." Access Token: ". $accessToken;
?>
```

respon.php



Invoice ID: J4y5wRMwvA8QYMlm2tzz Access Token:
zbswnMbUOASkgpgs6Szvhr2Jt4EMtPkg9iELNMG9Gwj4Whxfme

invoiceId
J4y5wRMwvA8QYMlm2tzz

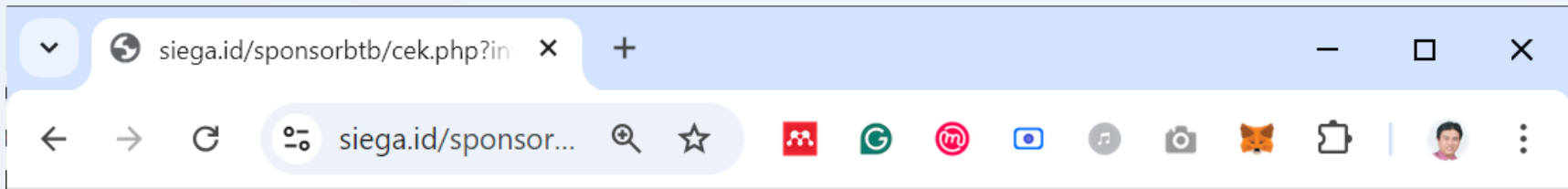
accessToken
zbswnMbUOASkgpgs6Szvhr2Jt4EMtPkg9iELNMG9Gwj4Whxfme

cek.php

```
<?php
include "aiyo_config.php";
$pathInvoice = '/api/v1/invoice';
$invoiceId = $_GET['invoiceId'];
$accessToken = $_GET['accessToken'];
$pathInvoice."/".$invoiceId."?accessToken=".$accessToken;
$URLCekStatus = $host . $pathInvoice."/".$invoiceId."?accessToken=".$accessToken;
$chCekInvoice = curl_init($URLCekStatus);
curl_setopt($chCekInvoice, CURLOPT_TIMEOUT, 30);
curl_setopt($chCekInvoice, CURLOPT_RETURNTRANSFER, TRUE);
$responseCekInvoice = curl_exec($chCekInvoice);
$cekInvoice=json_decode($responseCekInvoice);
$status=$cekInvoice->responseData->invoiceStatus;
echo "Invoice: ".$cekInvoice->responseData->invoiceName;
echo "<br/>Senilai: ".$cekInvoice->responseData->payAmount;
echo "<br/>Status: ".$status;
echo "<br/><br/><a href='\"".$cekInvoice->responseData->invoiceURL.\"' target='_blank'>Lanjutkan pembayaran</a>";
?>
```

Contoh Cek Status

- <https://siega.id/sponsorbtb/cek.php?invoiceId=J4y5wRMwvA8QYMIIm2tzz&accessToken=zbswnMbUOASkgpgs6Szvhr2Jt4EMtPkg9iELNMG9Gwj4Whxfme>



Invoice: Nama Komunitas

Senilai: 50000

Status: NEW

[Lanjutkan pembayaran](#)

Invoice For
Nama Komunitas

Amount
Rp. 50,000

Invoice Expired In
0 Days 02:42:28

Invoice Details

Reference ID: YPD0916095136

[NEW](#)

Name
Ridwan Sanjaya

Phone
0818000000

Email
ridwan@unika.ac.id

Invoice Date
16 September 2024

Note
-

Items	Price
Barang 1 (1)	Rp. 10,000
Barang 2 (2)	Rp. 20,000

Subtotal **Rp. 30,000**

Total Amount **Rp. 50,000**

Payment Method



! Payment method required!

CHECKOUT

Payment Menu

Invoice ID: J4y5wRMwvA8QYMI2tzz

QRIS QR Code Standar
Pembayaran Nasional



PT DUNIA BAYAR INDONESIA

Rp. 50,000

Gratis Biaya

CHECK STATUS

[Download QRIS](#)

🕒 Active Time

Expired In

23:44:53

Expired At

17 Sep 2024, 09.54.27 WIB

Contoh
Pembayaran

Contoh-Contoh Kode

- <https://bit.ly/PHPFinTech>

Thank You

ridwan@unika.ac.id